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REFRIGERATOR WITH ICE MAKER

Abstract

A refrigerator includes a freezing chamber and an ice making machine assembly. The ice making machine assembly includes an ice making machine main frame and an ice tray assembly. The refrigerator also includes a buckle structure that secures the ice making machine assembly to a top of the freezing chamber.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to Chinese Utility Model Patent Application No. CN 202420335549.5, filed on Feb. 22, 2024, entitled “Refrigerator with ice maker,” the entire disclosure of which is hereby incorporated by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to the technical field of household appliances, and in particular to a refrigerator with an ice maker.

BACKGROUND

[0003] Household refrigerators are used to store food at low temperatures within a containment space. The refrigerator is divided into a freezer compartment for storing food at a temperature below 0° C., and a refrigerator compartment for storing food at a temperature above 0° C. With improvement of living standards, refrigerators with ice machines are welcomed by more and more users. Ice machines can make ice cubes at home, which helps meet user demand for ice cubes, especially in summer.

[0004] At present, most ice machine components on the market are fixed on the refrigerator box through screws, which increases the material cost and assembly cost.

[0005] The technical problem to be solved by the present disclosure is to overcome the defects in the prior art that the ice making machine assembly is fixed on the refrigerator box through screws, resulting in high material and assembly costs. As set forth herein, the present disclosure provides a method that can reduce the material and assembly costs of refrigerators with ice makers.

SUMMARY

[0006] According to an aspect of the present disclosure, a refrigerator with an ice making machine includes a freezing chamber and an ice making machine assembly. The ice making machine assembly includes an ice making machine main frame and an ice making tray assembly. The ice tray assembly is installed in the main frame of the ice making machine and the main frame of the ice making machine is fixed on the top of the freezing chamber through a buckle structure.

[0007] According to another aspect of the present disclosure, a refrigerator includes a freezing chamber and an ice making machine assembly. The ice making machine assembly includes an ice making machine main frame and an ice tray assembly. The refrigerator also includes a buckle structure that secures the ice making machine assembly to a top of the freezing chamber.

[0008] According to another aspect of the present disclosure, a refrigerator includes a freezing chamber and an ice making machine assembly. The ice making machine assembly includes an ice making machine main frame and an ice tray assembly. The refrigerator also includes a buckle structure that secures the ice making machine assembly to a top of the freezing chamber. The buckle structure includes a plurality of mounting holes that are provided on a top of the ice making machine main frame. A plurality of hanging posts are disposed on the top of the freezing chamber. The plurality of hanging posts are engaged with the plurality of mounting holes.

[0009] The technical solution of this disclosure has several advantages. The refrigerator with an ice machine, as provided by the present disclosure, includes a construction where the main frame of the ice machine is fixed on the top of the freezing chamber through a buckle structure, the main frame of the ice machine and the freezing chamber are connected through buckles, and the connection method between the two is simple and convenient. This construction can reduce material and assembly costs through elimination of screw connections. This construction also facilitates the disassembly of the ice machine components for cleaning. At the same time, since the main frame of the ice making machine is fixed on the top of the freezing chamber, the freezing chamber provides the ice making capacity for the ice making machine assembly, and the length of the air duct that supplies cold air to the ice making machine assembly can be reduced.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] In order to more clearly explain the specific embodiments of the present disclosure or the technical solutions in the prior art, the drawings that need to be used in the description of the specific embodiments or the prior art will be briefly introduced below. Obviously, the drawings in the following description The accompanying drawings illustrate some embodiments of the present disclosure. For those of ordinary skill in the art, other drawings can be obtained based on these drawings without exerting creative efforts.

[0011] In the drawings:

[0012] FIG. **1** is a front elevational view of a refrigerator provided with an ice maker according to an embodiment of the present disclosure;

[0013] FIG. **2** is a side perspective schematic structural diagram of the ice making machine assembly shown in FIG. **1**;

[0014] FIG. **3** is a top perspective schematic structural diagram of the main frame of the ice machine shown in FIG. **2**;

[0015] FIG. **4** is a side elevational view of an ice machine assembly of the present disclosure;

[0016] FIG. **5** is a partial side elevational cross-sectional view of an ice machine assembly and box, after connection, according to the present disclosure;

[0017] FIG. **6** is a partial enlarged view of a connection feature of FIG. **5**; and

[0018] FIG. **7** is a side elevational schematic diagram of a cold air flow direction after an ice machine assembly is connected to a box according to the present disclosure.

[0019] The components in the figures are not necessarily to scale, emphasis instead being placed upon illustrating the principles described herein.

DETAILED DESCRIPTION

[0020] The present illustrated embodiments reside primarily in combinations of method steps and apparatus components related to an ice making machine of a refrigeration appliance. Accordingly, the apparatus components and method steps have been represented, where appropriate, by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present disclosure so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein. Further, like numerals in the description and drawings represent like elements.

[0021] For purposes of description herein, the terms “upper,” “lower,” “right,” “left,” “rear,” “front,” “vertical,” “horizontal,” and derivatives thereof shall relate to the disclosure as oriented in FIG. **1**. Unless stated otherwise, the term “front” shall refer to the surface of the element closer to an intended viewer, and the term “rear” shall refer to the surface of the element further from the intended viewer. However, it is to be understood that the disclosure may assume various alternative orientations, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

[0022] In the description of the present disclosure, it should be noted that, unless otherwise clearly stated and limited, the terms “installation”, “connection” and “connection” should be understood in a broad sense. For example, it can be a fixed connection or a removable connection. Detachable connection, or integral connection; it can be a mechanical connection or an electrical connection; it can be a direct connection or an indirect connection through an intermediate medium; it can be an internal connection between two components. For those of ordinary skill in the art, the specific meanings of the above terms in the present disclosure can be understood according to specific circumstances.

[0023] The terms “including,” “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises a . . . ” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element.

[0024] In addition, the technical features involved in different embodiments of the present disclosure described below can be combined with each other as long as they do not conflict with each other.

EXAMPLE

[0025] Household refrigerators are used to store food at low temperatures within a containment space. The refrigerator is divided into a freezer compartment for storing food at a temperature below 0° C. and a refrigerator compartment **3** for storing food at a temperature above 0° C.

[0026] At present, most ice machine components on the market are fixed on the refrigerator box through screws, which can increase the material cost and assembly cost.

[0027] In one embodiment, as shown in FIGS. **1-7**, an appliance includes a refrigeration compartment **3**, a freezer compartment **1**, and an ice maker assembly **2**. The ice maker assembly **2** includes an ice maker main frame **201** and an ice making tray **203**. The ice making tray **203** is installed in the main frame **201** of the ice maker assembly **2**, and the main frame **201** of the ice maker assembly **2** is fixed on the top of the freezing chamber **1** through a buckle structure.

[0028] In this embodiment, since the ice machine main frame **201** is fixed on the top of the freezing chamber **1** through a buckle structure, the ice machine main frame **201** and the freezing chamber **1** are connected through buckles. The connection method between the two is simple and convenient, and can reduce the size of the freezer. The material assembly costs caused by the absence of a screw connection make it easier to disassemble the ice maker assembly **2** for cleaning. Since the main frame **201** of the ice maker assembly **2** is fixed on the top of the freezing chamber **1**, the freezing chamber **1** provides the ice making capacity for the ice maker assembly **2**, and the length of the air duct that supplies cold air to the ice maker assembly **2** can be reduced.

[0029] With further reference to FIGS. **2-7**, the buckle structure includes a plurality of mounting holes **2011** provided on the top of the ice machine main frame **201** and a plurality of hanging posts **101** provided on the top of the freezing chamber **1**. Hanging posts **101** cooperate with the mounting hole **2011** to mount the ice maker assembly **2** on the top of the freezing chamber **1**. In this embodiment, the buckle structure includes a plurality of mounting holes **2011** and hanging posts **101** (FIGS. **5** and **6**). The hanging posts **101** cooperate with the mounting holes **2011** for hooking or securing the main frame **201** to the freezing chamber **1**. The connection between the main frame **201** of the ice maker assembly **2** and the freezing chamber **1** is simple and convenient. In an alternative embodiment, the mounting hole **2011** can be provided on the top of the freezing chamber **1** and the hanging post **101** can be provided on the top of the main frame **201** of the ice maker assembly **2**. In another alternative embodiment, a slot can be provided on the side of the main frame **201** of the ice maker assembly **2**, and an elastic buckle can be provided on a box **4** at the top of the freezing chamber **1**. The main frame **201** of the ice maker assembly **2** is fixed.

[0030] In one configuration, a refrigerator has a box body **4** (FIGS. **5** and **6**), a refrigerator compartment **3** and a freezing chamber **1** are formed in the box body **4**, and the hanging post **101** is located on the box body **4**.

[0031] In another configuration, as shown in FIGS. **2** and **3**, a total of four mounting holes **2011** are provided on the top of the ice machine main frame **201**, and correspondingly, four hanging posts

101 on the box **4** are also provided.

[0032] With reference again to FIG. **2**, the mounting hole **2011** has a first end **20111** facing the innermost side of the freezing chamber **1** and a second end **20112** facing the outside of the freezing chamber **1**. A width of the first end **20111** is greater than a width of the second end **20112**. The hanging post **101** includes a column **1011** that passes through the mounting hole **2011** and a first protruding portion **1012** protrudingly provided at the end of the column **1011**. An outer diameter of the first protruding portion **1012** is larger than the mounting hole **2011**. The width of the second end **20112** of the mounting hole **2011** is smaller than the width of the first end **20111** of the mounting hole **2011**. In this configuration, by making the width of the first end **20111** larger than the width of the second end **20112**, the column **1011** of the hanging post **101** can pass through the mounting hole **2011** and engage a first protrusion **1012** disposed at the end of the column **1011**. The outer diameter of the first protruding portion **1012** is greater than the width of the second end **20112** of the mounting hole **2011** and smaller than the width of the first end **20111** of the mounting hole **2011**. During installation, the column **1011** is first passed through the mounting hole **2011** at the first end **20111**, and then pushed toward or into the main frame **201** of the ice machine inward. The mounting hole **2011** moves relative to the column **1011**, and the first protruding portion **1012** is stuck on the lower side of the mounting hole **2011** near the second end **20112**, thereby preventing the ice machine main frame **201** from moving or falling downward. To disassemble the ice machine main frame **201**, the ice machine main frame **201** is first pulled outward and then the ice machine main frame **201** is moved downward so that the hanging post **101** separates from the mounting hole **2011**. Therefore, this embodiment facilitates the installation and disassembly of the ice machine main frame **201**.

[0033] Referring now to FIG. **3**, the installation hole **2011** includes a first hole section **20113** and a second hole section **20114** that are connected. The end of the first hole section **20113** is positioned away from the second hole section **20114** for installation. The first end **20111** of the hole **2011** is at a first end of the hanging plate **2012**. The second hole section **20114** is on a second end of the hanging plate **2012**. The first hole section **20113** is at the first end **20111** of the mounting hole **2011**. In the direction from the first end **20111** to the second end **20112**, the width of the first hole section **20113** gradually decreases and the width of the second hole section **20114** remains unchanged. In this configuration, since the width of the first hole section **20113** gradually decreases, when the column **1011** of the hanging post **101** moves in the installation hole **2011**, the moving direction can be guided. The width of the second hole section **20114** remains unchanged, and the hanging post **101** has a certain margin of movement in the second hole section **20114**, which can ensure that the main frame **201** of the ice maker assembly **2** will not fall off easily.

[0034] With further reference to FIG. **6**, the hanging post **101** further includes a second protruding portion **1013** protrudingly disposed outside the column **1011**. The second protruding portion **1013** and the first protruding portion **1012** extend axially away from the column **1011** and are spaced apart along a length of the column **1011**, and the distance between the first protruding portion **1012** and the second protruding portion **1013** matches or exceeds the depth of the mounting hole **2011**. With this construction, when the ice machine main frame **201** is installed in place, the second protruding portion **1013** is located on the upper side of the mounting hole **2011** and the first protruding portion **1012** is located on the lower side of the mounting hole **2011**. Thus, the second protruding portion **1013** and the first protruding portion **1012** sandwich the structure defining the mounting hole **2011** therebetween, which can further ensure that the ice maker main frame **201** is stably and reliably fixed in the freezer compartment **1**.

[0035] With further reference to FIG. **6**, the column **1011** is a hollow structure. Since the column **1011** has a hollow structure, it also has a certain degree of elasticity or flexibility, which facilitates the column **1011** to be clamped in the second hole section **20114**. Of course, it will be understood that the column **1011** may instead be a solid structure.

[0036] The outer diameter of the column **1011** may be slightly larger than the width of the second

hole section **20114**, such that the two frictionally engage and are connected through an interference fit.
[0037] With reference again to FIG. **2**, the top of the ice machine main frame **201** is provided with a hanging plate **2012** with a gap between the surface of the column **1011** and the mounting hole **2011** formed in the hanging plate **2012**. In this construction, the mounting holes **2011** are formed on the hanging plate **2012**, and the hanging plate **2012** is at a certain distance from the ice machine main frame **201**. Therefore, when the ice machine main frame **201** is installed in place, the top of the ice machine main frame **201** will include a gap between the surface and the top of the freezer **1** to facilitate air flow.
[0038] Referring further to FIGS. **2**, **3**, and **6**, an elastic stopper buckle **2013** is provided on the lower side of the installation hole **2011** close to a door of the freezing chamber **1** and, more specifically, close to a movable end of the door of the freezing chamber **1**. The elastic stopper buckle **2013** has an inclined plane **20131** that slopes toward the first end **20111** and a stop surface **20132** that is positioned toward the second end **20112**. When the hanging post **101** is clamped in the mounting hole **2011**, the stop surface **20132** prevents the hanging post **101** from moving within the mounting hole **2011** and toward the first end **20111**. During installation of the ice machine main frame **201**, when the elastic stopper buckle **2013** is provided, the column **1011** of the hanging post **101** is first passed through the first hole section **20113** and then the ice maker assembly **2** is pushed inward. The main body frame **201** and the hanging post **101** slide relatively along the inclined plane **20131**. When the hanging post **101** crosses the inclined plane **20131**, the hanging post **101** is located proximate the stop surface **20132**. The stop surface **20132** prevents the hanging post **101** from facing the first direction relative to the installation hole **2011**. When it is necessary to disassemble the main frame **201** of the ice maker assembly **2**, a user can press the elastic stopper buckle **2013** inward and pull the main frame **201** of the ice maker assembly **2** outward and downward so that the hanging post **101** it is separated from the mounting hole **2011**.
[0039] It should be noted that one end of the door of the freezing compartment **1** is hinged with the box **4** of the refrigerator appliance, and the other end of the door of the freezing compartment **1** is movable between open and closed positions.
[0040] With reference to FIG. **7**, the main frame **201** of the ice machine is provided with an induced air duct **2014** that is connected to the freezing air duct **5**. In this embodiment, the induced air duct **2014** can introduce the cold air in the freezing air duct **5** into the main frame **201** of the ice maker assembly **2**. An inclined baffle **2015** is provided at the outlet of the induced air duct **2014**. The inclined baffle **2015** is used to guide the cold air to the ice tray assembly to provide cooling capacity for ice making, as well as improved efficiency for ice making. This embodiment uses the main frame **201** of the ice machine to guide air without the need for additional air ducts, thus making the structure more simple and compact.
[0041] As shown in FIG. **2**, the air inlet of the induced air duct **2014** is located at one end of the ice machine main frame **201**. When the ice machine main frame **201** is installed in place, the induced air duct **2014** is in contact with the freezing air.
[0042] With reference now to FIGS. **4** and **5**, the ice tray assembly includes an ice making tray frame **202** and an ice making tray **203** installed in the ice making tray frame **202**. The ice making tray frame **202** is detachably inserted in the ice machine main frame **201**. The ice machine main frame **201** is connected with an ice turning motor **204** and an ice detection rod **205** (FIG. **2**). The ice turning motor **204** is connected with the ice making tray **203** and is used to drive the ice making tray **203** to flip. In this configuration, since the ice making tray frame **202** is detachably inserted into the ice making machine main frame **201**, the ice making tray frame **202** and the ice making tray **203** can be detached from the ice making machine main frame **201** to facilitate cleaning. The ice detector rod **205** is used to monitor whether the ice in the ice making tray **203** is full, and the ice turning motor **204** is used to turn over the ice making tray **203** when the ice in the ice making tray **203** is full so that the ice cubes in the ice making tray **203** fall off. Further, the inclined baffle **2015**

is used to guide the cold air to the ice making tray **203** of the ice tray assembly.

[0043] According to an aspect of the present disclosure, a refrigerator with an ice making machine includes a freezing chamber and an ice making machine assembly. The ice making machine assembly includes an ice making machine main frame and an ice making tray assembly. The ice tray assembly is installed in the main frame of the ice making machine and the main frame of the ice making machine is fixed on the top of the freezing chamber through a buckle structure.

[0044] According to another aspect of the present disclosure, a buckle structure includes a plurality of installation holes provided on the top of the main frame of the ice maker and a plurality of hanging posts provided on the top of the freezing compartment. The hanging posts cooperate with the installation holes.

[0045] According to another aspect of the present disclosure, a mounting hole has a first end facing the innermost side of the freezing chamber and a second end facing outside the freezing chamber. A width of the first end is greater than a width of the second end. A hanging post includes a cylinder passing through the mounting hole and a first protruding portion protrudingly provided at the end of the cylinder. The outer diameter of the first protruding portion is greater than the width of the second end of the mounting hole and smaller than the width of the second end of the mounting hole.

[0046] According to another aspect of the present disclosure, an installation hole includes a first hole section and a second hole section that are connected. An end of the first hole section is positioned away from the second hole section and the third hole section is positioned one end of the two hole segments away from the first hole segment. In the direction from the first end to the second end, the width of the first hole segment gradually decreases and the width of the second hole section remains unchanged.

[0047] According to another aspect of the present disclosure, a hanging column further includes a second protruding portion protrudingly disposed outside the column. The second protruding portion and the first protruding portion are spaced apart along an axial direction of the hanging column. A distance between the first protruding portion and the second protruding portion is adapted to the hole depth of the mounting hole.

[0048] According to another aspect of the present disclosure, the column is a hollow structure.

[0049] According to another aspect of the present disclosure, a hanging plate has a gap with the top surface of the ice making machine cylinder frame is provided on the top of the main frame of the ice machine, and the mounting hole is formed on the hanging plate.

[0050] According to another aspect of the present disclosure, an elastic stop buckle is provided on the lower side of the installation hole close to the freezing door and, more specifically, close to the movable end of the freezing door. The elastic stop buckle has a slope toward the first end and a stop surface at the second end. When the hanging post is clamped in the mounting hole, the stop surface prevents the hanging post from moving toward the first end relative to the mounting hole.

[0051] According to another aspect of the present disclosure, a main frame of the ice machine is provided with an induced air duct connected to the freezing air duct, and an inclined baffle is provided at the outlet of the induced air duct. The inclined baffle is used to guide the cold air to the ice tray assembly.

[0052] According to another aspect of the present disclosure, an ice making tray assembly includes an ice making tray frame and an ice making tray installed in the ice making tray frame. The ice making tray frame is detachably inserted into the ice making machine main body frame. The main frame of the ice making machine is connected with an ice turning motor and an ice detection rod. The ice turning motor is connected with the ice making tray and is used to drive the ice making tray to turn over.

[0053] According to another aspect of the present disclosure, a refrigerator includes a freezing chamber and an ice making machine assembly. The ice making machine assembly includes an ice making machine main frame and an ice tray assembly. The refrigerator also includes a buckle

structure that secures the ice making machine assembly to a top of the freezing chamber.
[0054] According to still another aspect of the present disclosure, a buckle structure includes a plurality of mounting holes provided on a top of an ice making machine main frame.
[0055] According to another aspect of the present disclosure, a plurality of hanging posts are disposed on the top of a freezing chamber. The plurality of hanging posts are engaged with a plurality of mounting holes.
[0056] According to yet another aspect of the present disclosure, each mounting hole of a plurality of mounting holes has a first end facing an innermost side of a freezing chamber. A second end of each mounting hole of the plurality of mounting holes is disposed outside of the freezing chamber. A width of the first end is greater than a width of the second end.
[0057] According to another aspect of the present disclosure, each hanging post of a plurality of hanging posts includes a hole that passes through a second end.
[0058] According to still another aspect of the present disclosure, an ice making machine assembly includes a hanging post that has a column. The column includes a first protruding portion provided at an end of the column. An outer diameter of the first protruding portion is larger than a mounting hole.
[0059] According to yet another aspect of the present disclosure, each mounting hole of a plurality of mounting holes includes a first hole section and a second hole section that are connected.
[0060] According to another aspect of the present disclosure, a first hole section includes a width that gradually decreases and a width of the second hole section remains unchanged.
[0061] According to still another aspect of the present disclosure, a hanging post includes a second protruding portion disposed outside a column.
[0062] According to another aspect of the present disclosure, a second protruding portion and a first protruding portion are arranged at intervals along an axial direction of a column.
[0063] According to yet another aspect of the present disclosure, a distance between a first protruding portion and a second protruding portion is commensurate with a depth of a mounting hole.
[0064] According to another aspect of the present disclosure, a refrigerator includes a freezing chamber and an ice making machine assembly. The ice making machine assembly includes an ice making machine main frame and an ice tray assembly. The refrigerator also includes a buckle structure that secures the ice making machine assembly to a top of the freezing chamber. The buckle structure includes a plurality of mounting holes that are provided on a top of the ice making machine main frame. A plurality of hanging posts are disposed on the top of the freezing chamber. The plurality of hanging posts are engaged with the plurality of mounting holes.
[0065] According to another aspect of the present disclosure, a refrigerator includes a hollow column. A top of the ice making machine main frame is provided with the hollow column. The top surface of the ice making machine main frame has a hanging plate with a gap. A plurality of mounting holes are formed on the hanging plate.
[0066] According to still another aspect of the present disclosure, a door is hingedly coupled to a freezing chamber and has a moveable end. An elastic stopper buckle is provided on a lower side of each mounting hole of a plurality of mounting holes close to the movable end of the door.
[0067] According to another aspect of the present disclosure, an elastic stopper buckle has an inclined surface that slopes toward a first end of a mounting hole and a stop surface disposed at a second end of the mounting hole. The stop surface prevents the hanging post from moving toward the first end relative to the mounting hole.
[0068] According to another aspect of the present disclosure, an ice making machine main frame includes an air guide connected to a freezing air duct. An inclined baffle is disposed at an outlet of an induced air duct and is used to guide cold air to an ice tray assembly.
[0069] According to another aspect of the present disclosure, an ice tray assembly includes an ice making tray frame and an ice making tray that is installed on the ice making tray frame. The ice

making tray frame is detachably inserted into the ice making machine main frame.
[0070] According to yet another aspect of the present disclosure, an ice making machine main frame is connected with an ice turning motor and an ice detection rod. The ice turning motor is connected to the ice making tray and drives the ice making tray to turn over.
[0071] It will be understood by one having ordinary skill in the art that construction of the described disclosure and other components is not limited to any specific material. Other exemplary embodiments of the disclosure disclosed herein may be formed from a wide variety of materials, unless described otherwise herein.
[0072] For purposes of this disclosure, the term “coupled” (in all of its forms, couple, coupling, coupled, etc.) generally means the joining of two components (electrical or mechanical) directly or indirectly to one another. Such joining may be stationary in nature or movable in nature. Such joining may be achieved with the two components (electrical or mechanical) and any additional intermediate members being integrally formed as a single unitary body with one another or with the two components. Such joining may be permanent in nature or may be removable or releasable in nature unless otherwise stated.
[0073] It is also important to note that the construction and arrangement of the elements of the disclosure as shown in the exemplary embodiments is illustrative only. Although only a few embodiments of the present innovations have been described in detail in this disclosure, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter recited. For example, elements shown as integrally formed may be constructed of multiple parts or elements shown as multiple parts may be integrally formed, the operation of the interfaces may be reversed or otherwise varied, the length or width of the structures and/or members or connector or other elements of the system may be varied, the nature or number of adjustment positions provided between the elements may be varied. It should be noted that the elements and/or assemblies of the system may be constructed from any of a wide variety of materials that provide sufficient strength or durability, in any of a wide variety of colors, textures, and combinations. Accordingly, all such modifications are intended to be included within the scope of the present innovations. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions, and arrangement of the desired and other exemplary embodiments without departing from the spirit of the present innovations.
[0074] It will be understood that any described processes or steps within described processes may be combined with other disclosed processes or steps to form structures within the scope of the present disclosure. The exemplary structures and processes disclosed herein are for illustrative purposes and are not to be construed as limiting.

Claims

1-10. (canceled)
11. A refrigerator, comprising: a freezing chamber; an ice making machine assembly, wherein the ice making machine assembly includes an ice making machine main frame and an ice tray assembly; and a buckle structure that secures the ice making machine assembly to a top of the freezing chamber.
12. The refrigerator of claim 11, wherein the buckle structure includes a plurality of mounting holes provided on a top of the ice making machine main frame.
13. The refrigerator of claim 12, further comprising: a plurality of hanging posts disposed on the top of the freezing chamber, wherein the plurality of hanging posts are engaged with the plurality of mounting holes.

14. The refrigerator of claim 13, wherein each mounting hole of the plurality of mounting holes has a first end facing an innermost side of the freezing chamber.
15. The refrigerator of claim 14, wherein a second end of each mounting hole of the plurality of mounting holes is disposed outside of the freezing chamber, and wherein a width of the first end is greater than a width of the second end.
16. The refrigerator of claim 15, wherein each hanging post of the plurality of hanging posts includes a hole that passes through the second end.
17. The refrigerator of claim 12, wherein the ice making machine assembly includes a hanging post having a column, wherein the column includes a first protruding portion provided at an end of the column, and wherein an outer diameter of the first protruding portion is larger than the mounting hole.
18. The refrigerator of claim 12, wherein each mounting hole of the plurality of mounting holes includes a first hole section and a second hole section that are connected.
19. The refrigerator of claim 18, wherein the first hole section includes a width that gradually decreases and a width of the second hole section remains unchanged.
20. The refrigerator of claim 17, wherein the hanging post includes a second protruding portion disposed outside the column.
21. The refrigerator of claim 20, wherein the second protruding portion and the first protruding portion are arranged at intervals along an axial direction of the column.
22. The refrigerator of claim 21, wherein a distance between the first protruding portion and the second protruding portion is commensurate with a depth of the mounting hole.
23. A refrigerator, comprising: a freezing chamber; an ice making machine assembly, wherein the ice making machine assembly includes an ice making machine main frame and an ice tray assembly; a buckle structure that secures the ice making machine assembly to a top of the freezing chamber, wherein the buckle structure includes a plurality of mounting holes provided on a top of the ice making machine main frame; and a plurality of hanging posts disposed on the top of the freezing chamber, wherein the plurality of hanging posts are engaged with the plurality of mounting holes.
24. The refrigerator of claim 23, further comprising: a hollow column.
25. The refrigerator of claim 23, wherein the top of the ice making machine main frame is provided with the hollow column, and wherein the top surface of the ice making machine main frame has a hanging plate with a gap, and further wherein the plurality of mounting holes are formed on the hanging plate.
26. The refrigerator of claim 25, further comprising: a door hingedly coupled to the freezing chamber, the door having a moveable end; and an elastic stopper buckle provided on a lower side of each mounting hole of the plurality of mounting holes close to the movable end of the door.
27. The refrigerator of claim 26, wherein the elastic stopper buckle has an inclined surface that slopes toward a first end of the mounting hole and a stop surface disposed at a second end of the mounting hole, and wherein the stop surface prevents the hanging post from moving toward the first end relative to the mounting hole.
28. The refrigerator of claim 23, wherein the ice making machine main frame includes an air guide connected to a freezing air duct, and wherein an inclined baffle is disposed at an outlet of an induced air duct, the inclined baffle used to guide cold air to the ice tray assembly.
29. The refrigerator of claim 23, wherein the ice tray assembly includes an ice making tray frame and an ice making tray installed on the ice making tray frame, and wherein the ice making tray frame is detachably inserted into the ice making machine main frame.
30. The refrigerator of claim 29, wherein the ice making machine main frame is connected with an ice turning motor and ice detection rod, and wherein the ice turning motor is connected to the ice making tray and drives the ice making tray to turn over.